



CSCE 2501

Fundamentals of Database Systems

Project - Data.gov

Prof. Hossam Sharara
Dept. of Computer Science, AUC

Introduction

In this project, you will be building a database application for US public governmental datasets from data.gov, that allows users to get information about the available public datasets, their types, publishing organization, metadata and tags.

The main site that provide the information is <https://catalog.data.gov/dataset/>. The website doesn't provide filtering or higher level aggregations for statistics. Hence, your application will be used to provide more flexibility on that front.

Your task will to build a database system backend and an application that provide the categorization, statistics and features described next for the first 2000 available datasets.

Milestone I: Data Requirements

The database system you design should be based on the information provided on the datasets in the first 100 pages, along with the associated organization. The URLs for each page of the datasets can be constructed by replacing the page number in this url (https://catalog.data.gov/dataset/?page=<page_number>) e.g. for the 1st page, the <page_number> should be replaced by “1”, etc. till the 100th page.

Your database design should store all the information about the dataset, including but not limited to: dataset name, description, access level, license, metadata creation and update dates, publisher, maintainer, identifier, topic (if available), associated tags, etc. For each dataset, you need to store the associated publishing organization, including its description, organization type, and contact information. For each dataset, you are required to also store the available file formats and the corresponding URLs.

The system should also allow users to register on the system using their email addresses, and pick a username, gender, age, birthdate, and country. After registering, the users can record their usage of any of the datasets along with the project name they used it for, the project category (analytics, machine learning, or field research).

Milestone I: Database Design & Implementation

Tools: MySQL

Deadline: 8th March, 2026

Description:

In this milestone, you are required to design the database ERD, create your database and schema in MySQL server for the data requirements presented on the previous slides.

Deliverables: (20 points)

- Entity-Relationship Diagram of the system
- Relational Model for your system
- A transcript for the SQL statements used to create your database and schema

Milestone II: Database Population

In this milestone, you are required to write a web crawler to crawl the pages from Data.gov for all the datasets in the first 100 pages, parse the HTML you crawl and extract the relevant fields for populating the non-user tables in your schema. Note that you may need to extract some of the information from the about pages for the associated organizations

At the end of this milestone, you should have all your non-user tables populated with the extracted data. For the user tables, use the included users file to populate the users table, and generate a random 500 entries for the usage information from the datasets crawled

Milestone II: Web Crawling and Data Population

Tools: Scrappy, pycharm, BeautifulSoup, selenium, python (or any other crawler library / language)

Deadline: 29th March, 2026

Description:

In this milestone, you are required to implement your crawler, extract the data from the Wikipedia website and populate your database

Deliverables: (50 points)

- Crawling script
- Populated MySQL database dump
- CSV files for the values in each table

Milestone III: Application Layer

In the final milestone, you are required to design a client application that is capable of connecting to the database hosted on remote MySQL server. The application should have the following functionalities:

- Register a user
- Add a new user usage for a given dataset
- View existing usage information for the user
- View the datasets by the organization type (federal, state, city, etc.)
- View the top 5 contributing organizations by the number of datasets provided
- View the datasets available in a given format
- View the datasets associated with a given input tag
- Show the total number of datasets contributed by organization, topic, format, and organization type
- Show the top 5 dataset by number of users who are using them
- Show the distribution of the datasets usage by project type
- Display the top 10 tags associated with every project type

Milestone III: Application Layer

Tools: Any language / platform to implement your application (Web-based, GUI, command-line are all acceptable)

Deadline: 19th April, 2026

Description:

In this milestone, you are required to implement an application to perform various transactions / queries on your backend database system. The database has to be hosted on an actual server (you can use any of the free DB hosting services available (e.g. www.db4free.net)) and make sure your application is communicating with this service not the localhost

Deliverables: (30 points + 5 bonus points for GUI/Web-based apps)

- Application Implementation / Source Code + Executable
- Latest Dump of your database
- Video Demo